

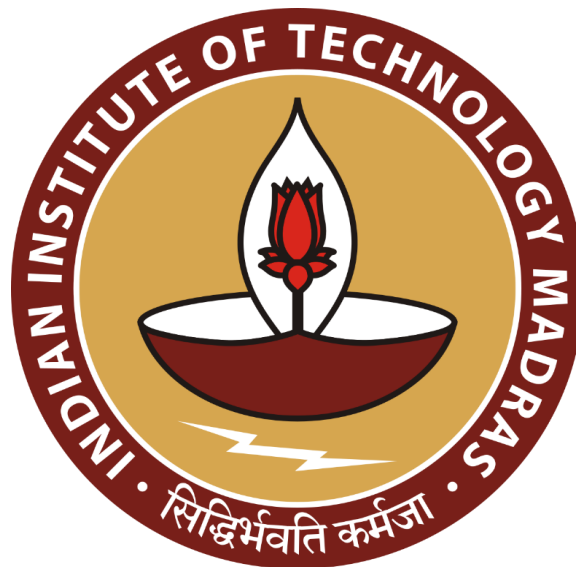
Optimizing Inventory Management and Product Profitability for Aastha RCM PUC

Mid-term Submission for the BDM capstone Project

Submitted by

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1 Executive Summary

Aastha RCM PUC, a department store located in Bero, Ranchi (Jharkhand), was established in October 2021 and operates under the sole proprietorship of Mrs. Sarika Devi. The store offers a range of products, including groceries, apparel, personal care, health supplements, and more under a single brand. While it has built strong customer loyalty and brand recognition, the store faces challenges in managing its inventory efficiently. Key issues include overstocking, stockouts, lack of demand forecasting, and limited visibility into product-wise profitability.

To address these challenges, we utilized 21 weeks of data from May 2024 to September 2024. The dataset includes critical columns such as Product Name, Opening Stock, Quantity In, Quantity Out, Cost Amount, Selling Amount, and Closing Balance. Techniques such as coefficient of variation-based demand fluctuation analysis and demand forecasting using simple moving average for grocery, food and health supplement products were used to identify demand patterns and improve stock planning. Microsoft Excel was used for data preprocessing, analysis, and visualization.

This mid-term report includes data source, metadata and descriptive statistics, analysis processes and methods and the results and findings. Results show that 75% of products show moderate demand fluctuation, 22% show stable demand fluctuation, and 3% show high fluctuation. Both the categories, Groceries and Food & Health Supplements have consistent growth trends. The final report will incorporate demand forecasting of all categories of products, further inventory management methods like safety stock and reorder point calculations, ABC classification and product profitability Analysis, and conclude with actionable recommendations for improving profitability and inventory optimization.

2 Proof of Originality of the Data

Details:

Name: Aastha RCM PUC

Founder/Owner: Mrs. Sarika Devi

Address: Tetartoli, Bero, Ranchi, Jharkhand(835202)

I have taken images of the shop and recorded an interview with the owner of the shop as proof of originality. In addition, I asked the owner to provide me a letter with stamp and sign as a no objection certificate.

Letter from the organization:

<https://drive.google.com/file/d/1FWBQIFSP70X82wxtaJ8sqP2TryWC-dw/view?usp=sharing>

Images of the organisation:

https://drive.google.com/drive/folders/1QhoFmOENVGOR2AeR_R71EXecLobszwBS?usp=sharing

Video interaction with the owner:

https://drive.google.com/file/d/1fWh6yA0jAm0XFi3hVb5H_hG7UzK67KZV/view?usp=sharing

3 Metadata and Descriptive Statistics

The department store Aastha RCM PUC has POS(point of sale), a computer-based system that allows businesses to keep track of sales and inventory data of each product of the store. So, excel files having compiled report of weekly data which includes comprehensive records related to inventory movement, sales, and stock status was collected for the time period of 21 weeks from May 2024-September 2024 directly from the system.

The Original excel data sheets contain:

Hsn Code: Unique code for each product

Product Name: The name or label used to identify a specific item

Op. Stock: The quantity of an item available in inventory at the beginning of the week

Qty In: The number of units received into stock during a week, representing restocking

Qty Out: The number of units sold from inventory during a week

Tax Rate: Applicable tax percentage

Amount: The total purchasing cost for an item, typically calculated as: $\text{Quantity In} * \text{Cost Price per Unit}$

Sell Amount: The total revenue for an item, calculated as: $\text{Quantity Out} * \text{Selling Price per Unit}$

Balance: The closing stock after accounting for 'Qty In' and 'Qty Out' transactions

The Structured data contains:

Product Name: The name or label used to identify an specific item

Opening Stock: The quantity of an item available in inventory at the beginning of the week

Quantity In: The number of units received into stock during a week, representing restocking

Quantity Out: The number of units sold from inventory during a week

Balance: The closing stock after accounting for 'Quantity In' and 'Quantity Out' transactions

Selling price per Unit: The price at which one unit of the product is sold

Cost price per Unit: The purchase cost of one unit of the product

Cost Amount: The total purchasing cost for an item, typically calculated as: $\text{Quantity In} * \text{Cost Price per Unit}$

Sell Amount: The total revenue for an item, calculated as: $\text{Quantity Out} * \text{Selling Price per Unit}$

Profit: The total profit in a week, calculated as $(\text{Selling Price per Unit} - \text{Cost Price per Unit}) * \text{Quantity Out}$.

Overall Dataset Statistics:

Metric	Op. Stock	Qty In	Qty Out	Cost Amount	Sell Amount	Profit
Sum	109805	156160	147986	8110037	8675419	1164145
Mean	145	207	196	10728	11475	1540
Median	94	91	71	3071	2803	317
Std. Dev.	161.97	365.29	359.06	26997.31	32851	6144.51
Min	0	0	0	0	0	0
Max	860	3001	3033	262902	321266	62432

Table 3.1: Descriptive statistics of all the columns of structured dataset

4 Analysis Processes and Methods

Data Preparation and Cleaning:

Data was collected in the form of Excel sheets with a sales and inventory data report, and the rows were deleted where most of the column values were zero, those are considered non-informative for the analysis. Some columns like tax rate were also deleted. Then, we organized the data into separate sheets for calculation and analysis and renamed columns for clarity.

Metadata and Descriptive Statistics:

Generated metadata for structured data and summarized the data by calculating descriptive statistics such as sum, mean, median, etc., for the Opening stock, Quantity In, Quantity Out, Cost Amount, Sell amount, and Profit columns.

Demand Fluctuation Analysis:

For each product, weekly sales data (Quantity Out) was used as the basis for analysis. Sales values were arranged in a format where each product's sales were tracked over 21 weeks. Coefficient of Variation (CV%) was used as the primary metric. To compute CV%, Average Sales and Standard Deviation were calculated for each product over 21 weeks, and then CV% was calculated as

$$CV\% = (\text{Standard deviation} / \text{Mean}) * 100$$

Here, the Standard Deviation measures the amount of variation or dispersion in sales data, and the Mean is the average sales value.

This metric expresses demand fluctuation relative to the average, allowing for fair comparisons across

products with different sales volumes.

Then, the products were classified into three demand fluctuation categories:

Stable Demand: $CV\% \leq 50\%$

Moderate Fluctuation: $50\% < CV\% \leq 100\%$

High Fluctuation: $CV\% > 100\%$

CV%-based analysis enables businesses to design inventory strategies based on product demand behavior. Stable-demand items can be managed with fixed reorder points and minimal safety stock, while fluctuating-demand items require dynamic stock levels and regular review. This approach reduces overstock and stockout risks and thus enhances inventory optimization.

A classification formula was applied in Excel using IF() logic to assign categories to each product.

Then, a summary table was created showing the count of products in each category and a pie chart was generated to represent the proportional distribution of products by demand type. We are using a pie chart to represent the distribution of products based on demand fluctuation categories because it is ideal for visualizing proportional relationships in a dataset than any other graph.

Demand Forecasting with Moving Average:

This analysis was done to understand weekly sales trends for different categories of products and to forecast near-future demand. The Simple Moving Average (SMA) method was selected to achieve this. The SMA is computed by taking the average of the actual sales(Qty Out) over a fixed number of past weeks. This average serves as the forecast for the next week. We calculated SMA by taking the average sales quantity of the previous three weeks for each product, starting from Week 4, and extending it up to Week 22, as follows

$$SMA(Week\ 22) = (Sales(Week\ 19) + Sales(Week\ 20) + Sales(Week\ 21)) / 3$$

For week 23, the SMA was calculated using actual sales of weeks 20, 21, and the forecasted value of week 22. Similarly, for week 24, the average was taken of week 21 sales, week 22 forecast, and week 23 forecast, as follows

$$SMA(Week\ 23) = (Sales(Week\ 20) + Sales(Week\ 21) + SMA(Week\ 22)) / 3$$

$$SMA(Week\ 24) = (Sales(Week\ 21) + SMA(Week\ 22) + SMA(Week\ 23)) / 3$$

The Simple Moving Average method was chosen because it allows for quick and practical insights into a realistic trend line that could be used to forecast short-term demand and make better inventory and ordering decisions without requiring complex statistical techniques. Forecasting future demand helps predict how much stock will be needed in the upcoming weeks. By accurately projecting demand for each product, stockouts can be minimized because the business can order or stock up in advance based on forecasted sales, especially for high-demand products. Also, overstock situations are reduced, as forecasting avoids unnecessary inventory buildup for low-demand items, which can tie up capital and

storage space.

We are using line charts for visualization because they clearly show trends over time, patterns, fluctuations, and direction of change at a glance.

Tools used:

Microsoft Excel was used to organize and preprocess raw data, structure it into a spreadsheet format, perform calculations using Excel formulas, and create visual representations such as bar charts, pie charts, and line plots for further analysis.

5 Results and Findings

The Demand Fluctuation analysis findings showed that:

75% of the products fell under Moderate Fluctuation, indicating sales variability that requires dynamic inventory planning and safety stock, 22% of the products had Stable Demand, suitable for fixed reorder levels and consistent stocking and only 3% of the products showed High Fluctuation, demanding special attention due to unpredictable sales. These insights will enable the business to tailor inventory strategies by demand type, improving stock efficiency and minimizing risks of overstocking or stockouts.

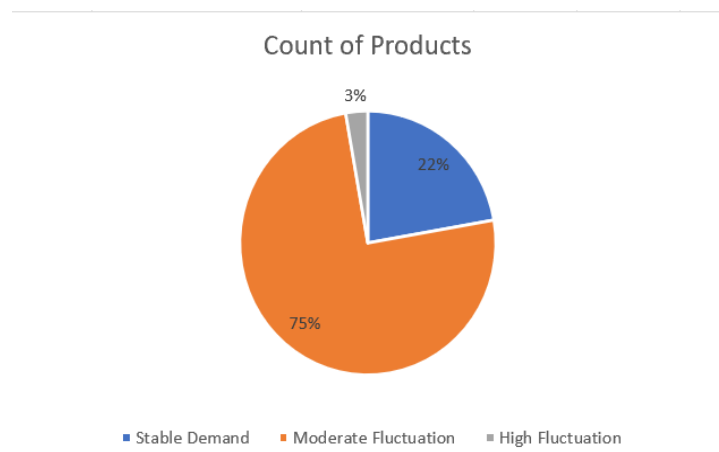


Figure 5.1: Distribution of products based on demand fluctuation

The pie chart shown in figure 5.1 represents the distribution of products based on demand fluctuation categories which are Stable Demand, Moderate Fluctuation, and High Fluctuation.

Demand Forecasting:

Demand forecasting was done for all the categories of products which are Grocery, Food & Health Supplements, Personal & Home Care products, and Apparel and Footwear. The first two categories are included in this report.

Demand Forecasting for Week 22-24 for the products of Grocery category:

The last three SMA values are the demand forecasting for the week 22, week 23 and week 24.

Week no.	Grocery									
	Masoor Malka(Qty Out)	Masoor Malka(SMA)	Tea(Qty Out)	Tea(SMA)	Chilli Powder(Qty Out)	Chilli Powder(SMA)	Coriander Powder(Qty Out)	Coriander Powder(SMA)	Turmeric Powder(Qty Out)	Turmeric Powder(SMA)
1	9	-	24	-	6	-	7	-	32	-
2	18	-	34	-	15	-	18	-	47	-
3	12	-	44	-	18	-	13	-	76	-
4	35	13	46	34	35	13	37	13	69	52
5	35	22	50	41	32	23	32	23	81	64
6	14	27	49	47	4	28	14	27	80	75
7	13	28	52	48	3	24	14	28	85	77
8	26	21	76	50	18	13	21	20	98	82
9	40	18	108	59	22	8	31	16	129	88
10	29	26	82	79	6	14	21	22	108	104
11	49	32	117	89	37	15	41	24	151	112
12	63	39	123	102	39	22	56	31	154	129
13	73	47	121	107	97	27	67	39	150	138
14	53	62	162	120	30	58	43	55	192	152
15	54	63	117	135	52	55	45	55	153	165
16	72	60	220	133	52	60	60	52	263	165
17	85	60	161	166	64	45	110	49	214	203
18	97	70	171	166	58	56	90	72	225	210
19	91	85	184	184	46	58	93	87	226	234
20	117	91	230	172	72	56	123	98	264	222
21	112	102	214	195	40	59	118	102	252	238
22		107		209		53		111		247
23		112		218		55		117		254
24		110		214		49		116		251

Table 5.1: Actual Sales and Calculated SMA of different products of Grocery category

Garam Masala(Qty Out)	Garam Masala(SMA)	Sabji Masala(Qty Out)	Sabji Masala(SMA)	Atta(Qty Out)	Atta(SMA)	Rice Bran Oil(Qty Out)	Rice Bran Oil(SMA)	Salt(Qty Out)	Salt(SMA)
16	-	44	-	17	-	219	-	134	-
25	-	59	-	26	-	270	-	225	-
33	-	79	-	36	-	365	-	218	-
51	25	110	61	23	26	415	285	266	192
59	36	152	83	24	28	591	350	314	236
33	48	137	114	24	28	588	457	340	266
33	48	146	133	24	24	615	531	367	307
34	42	169	145	32	24	705	598	440	340
51	33	189	151	62	27	802	636	614	382
37	39	190	168	35	39	854	707	690	474
69	41	216	183	69	43	931	787	572	581
74	52	270	198	70	55	982	862	885	625
132	60	267	225	64	58	1037	922	690	716
65	92	250	251	106	68	1064	983	643	716
89	90	257	262	67	80	1142	1028	725	739
89	95	279	258	174	79	1200	1081	712	686
106	81	316	262	103	116	1387	1135	809	693
101	95	329	284	122	115	1522	1243	825	749
94	99	341	308	112	133	1456	1370	864	782
125	100	390	329	135	112	1671	1455	972	833
92	107	398	353	124	123	1718	1550	1043	887
	104		376		124		1615		960
	107		388		128		1668		992
	101		387		125		1667		998

Table 5.2: Actual Sales and Calculated SMA of different products of Grocery category

Line Charts of Actual Sales vs. Simple Moving Average for Grocery Category Products:

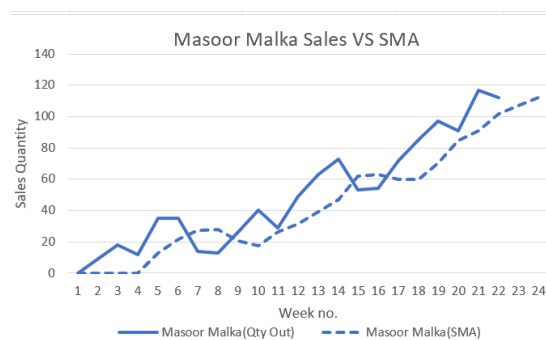


Figure 5.2: Masoor Malka Actual Sales VS SMA

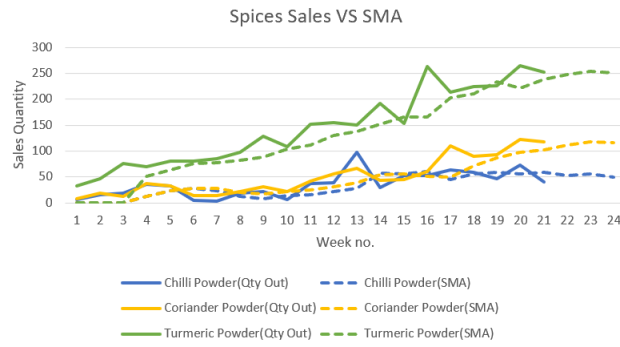


Figure 5.3: Spices Actual Sales VS SMA

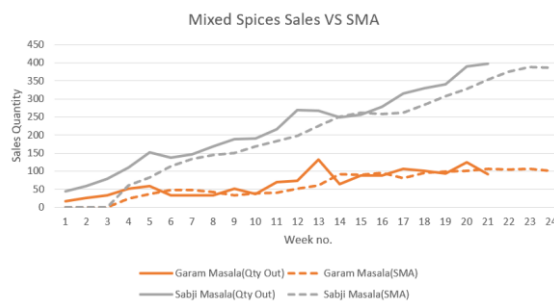


Figure 5.4: Mixed Spices Actual Sales VS SMA

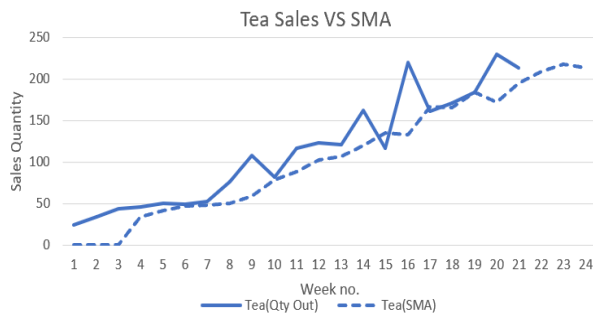


Figure 5.5: Tea Actual Sales VS SMA

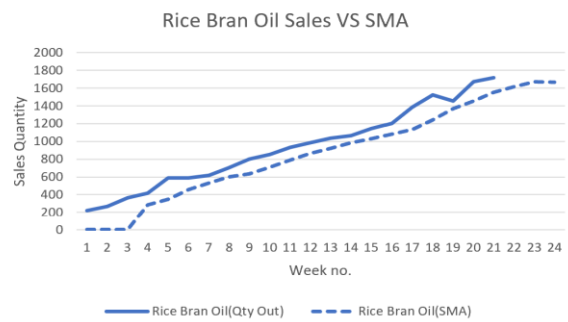


Figure 5.6: Rice Bran Oil Actual Sales VS SMA

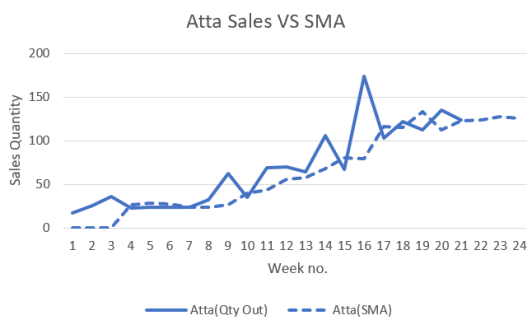


Figure 5.7: Atta Actual Sales VS SMA

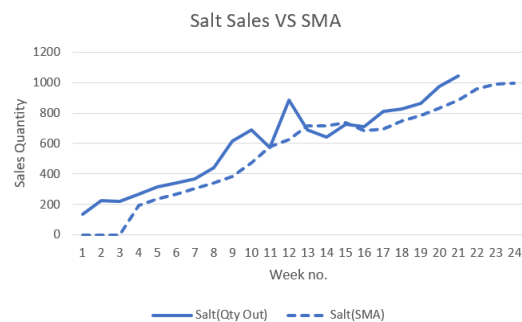


Figure 5.8: Salt Actual Sales VS SMA

The grocery category shows a clear upward sales trend across everyday items. Products like Masoor Maska, Tea, and Rice Bran Oil continue to grow steadily, with Rice Bran Oil showing especially strong growth. Atta and Salt also see rising trends, with small ups and downs. Among spices, Chilli Powder shows steady increases, while Coriander and Turmeric Powders have more noticeable shifts. Mixed spices like Garam Masala maintain steady sales, but Sabji Masala is growing quickly and appears to be in high demand. The simple moving average (SMA) lines help smooth out short-term fluctuations and make it easier to see the overall direction of customer demand over time.

Demand Forecasting for Week 22-24 for the products of Food and Health Supplement categories:

The last three SMA values are the demand forecasting for the week 22, week 23 and week 24.

Week	Food and Health Supplement									
	Noodles(Qty Out)	Noodles(SMA)	Fiber Biscuit(Qty Out)	Fiber Biscuit(SMA)	Jeera Biscuit(Qty Out)	Jeera Biscuit(SMA)	Mixed Pickle(Qty Out)	Mixed Pickle(SMA)	Ketchup(Qty Out)	Ketchup(SMA)
1	82	-	13	-	19	-	9	-	11	-
2	105	-	21	-	32	-	18	-	19	-
3	126	-	32	-	34	-	10	-	29	-
4	160	104	20	22	58	28	35	12	17	20
5	203	130	21	24	61	41	28	21	15	22
6	208	163	22	24	36	51	7	24	14	20
7	343	190	28	21	42	52	6	23	18	15
8	310	251	40	24	61	46	11	14	19	16
9	360	287	67	30	82	46	18	8	46	17
10	377	338	49	45	68	62	7	12	21	28
11	419	349	82	52	115	70	26	12	54	29
12	555	385	81	66	127	88	41	17	54	40
13	519	450	75	71	151	103	52	25	44	43
14	506	498	120	79	117	131	28	40	86	51
15	516	527	75	92	139	132	29	40	40	61
16	557	514	186	90	143	136	44	36	144	57
17	664	526	126	127	175	133	55	34	71	90
18	727	579	137	129	171	152	65	43	79	85
19	749	649	144	150	167	163	52	55	68	98
20	892	713	165	136	202	171	71	57	94	73
21	928	789	151	149	170	180	67	63	74	80
22		856		153		180		63		79
23		892		156		184		67		82
24		892		154		178		66		78

Table 5.3: Actual Sales and Calculated SMA of different products of Food and Health Supplement category

Gooddot Protein(Qty Out)	Gooddot Protein(SMA)	Nutricharge Woman(Qty Out)	Nutricharge Woman(SMA)	Nutricharge Veg Omega(Qty Out)	Nutricharge Veg Omega(SMA)	Nutricharge Man(Qty Out)	Nutricharge Man(SMA)	Mixture(Qty Out)	Mixture(SMA)	Bhujia(Qty Out)	Bhujia(SMA)
16	-	27	-	13	-	17	-	8	-	14	-
27	-	38	-	22	-	28	-	18	-	24	-
34	-	36	-	32	-	35	-	11	-	36	-
52	26	63	34	20	22	55	27	37	12	38	25
53	38	65	46	26	25	57	39	32	22	52	33
28	46	49	55	25	26	30	49	14	27	58	42
37	44	55	59	26	24	35	47	13	28	62	49
57	36	70	56	33	26	42	41	20	20	74	57
51	31	83	58	61	28	62	36	26	16	104	65
83	41	83	69	37	40	52	46	20	20	84	80
89	48	104	79	70	44	85	52	41	22	116	87
149	64	121	90	70	56	90	66	56	29	127	101
86	74	139	103	61	59	151	76	68	39	117	109
111	107	117	121	102	67	86	109	44	55	163	120
115	108	123	126	57	78	113	109	46	56	117	136
138	115	301	126	163	73	119	117	59	53	223	132
148	104	163	180	89	107	137	106	70	50	216	168
144	121	181	196	99	103	134	123	83	58	227	185
180	134	175	215	88	117	122	130	71	71	235	222
147	143	197	173	108	92	143	131	93	75	260	226
	157	196	184	90	98	114	133	90	82	259	241
	157		189		95		126		85		251
	161		194		98		128		89		257
	155		193		94		123		88		256

Table 5.4: Actual Sales and Calculated SMA of different products of Food and Health Supplement category

Line Charts of Actual Sales vs. Simple Moving Average for Food and Health Supplement Categories Products:

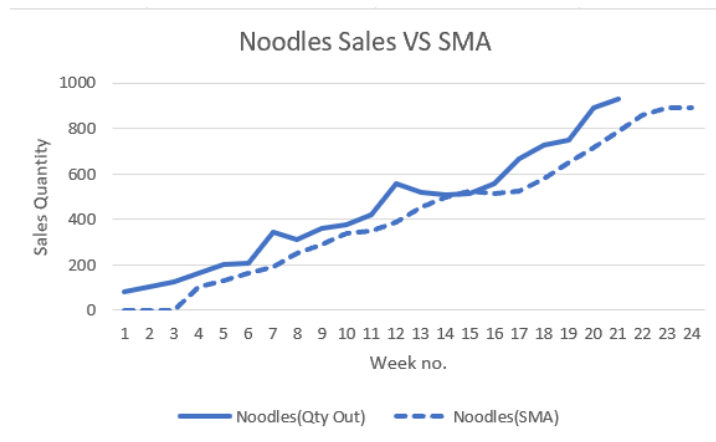


Figure 5.9: Noodles Actual Sales VS SMA

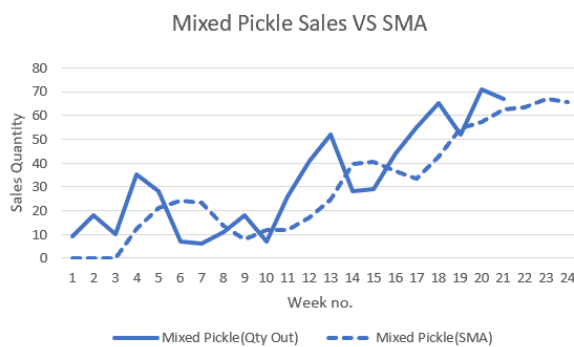


Figure 5.10: Mixed Pickle Sales VS SMA

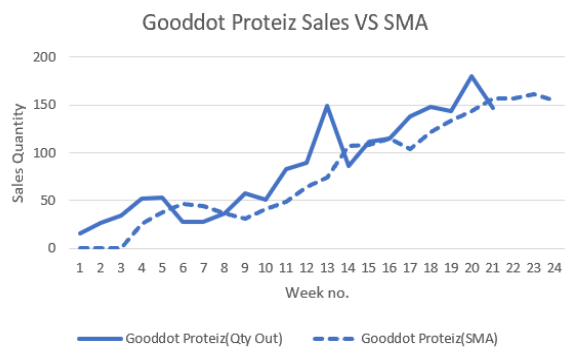


Figure 5.11: Gooddot Proteiz Sales VS SMA

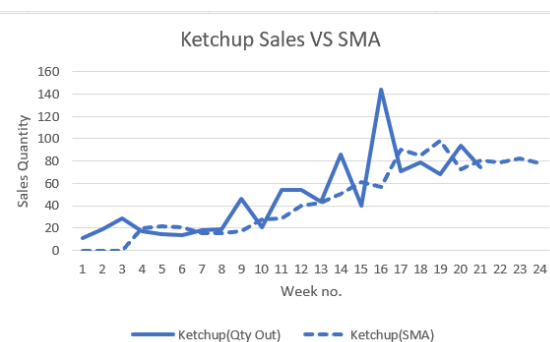


Figure 5.12: Ketchup Sales VS SMA

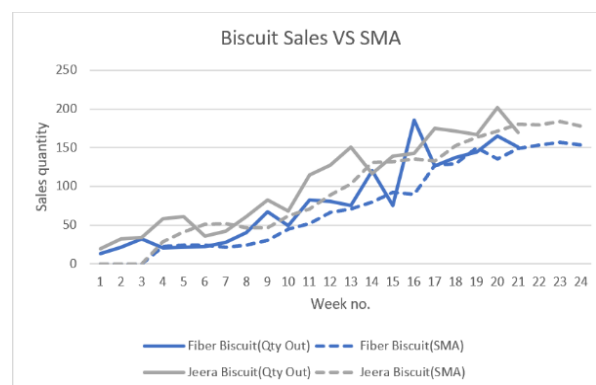


Figure 5.13: Biscuit Sales VS SMA

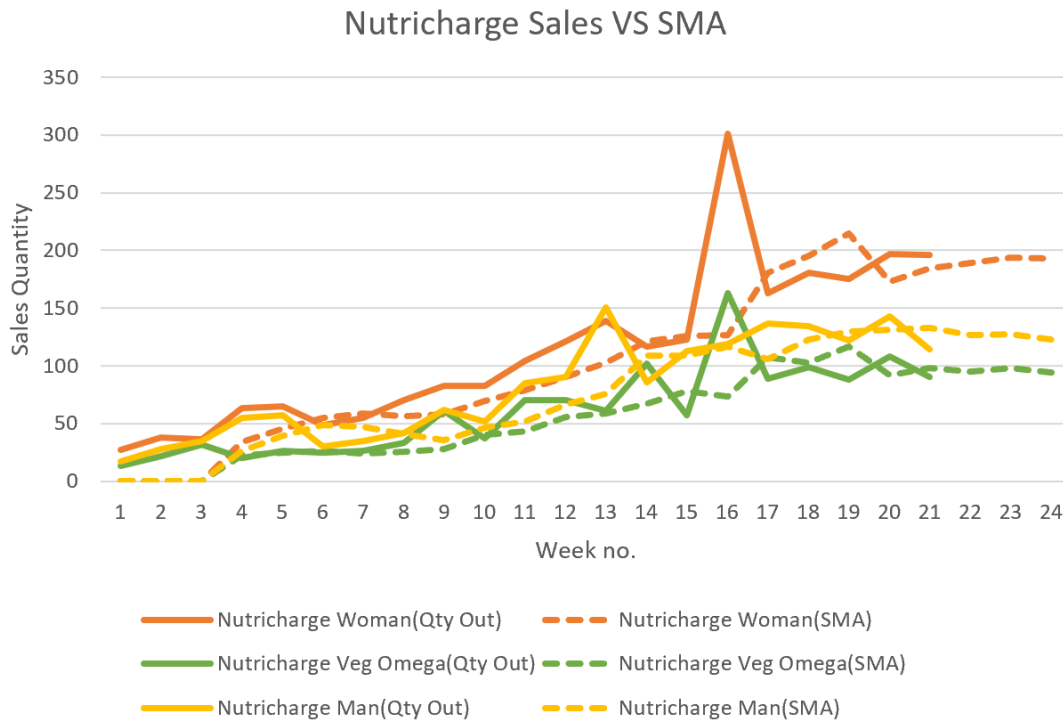


Figure 5.14: Actual Sales VS SMA of Nutrcharge products

This category also shows increasing customer interest, both in easy-to-use and health-related products. Noodles have a steady, straight-line growth, showing consistent popularity. Biscuits, especially Jeera and Snacks like Bhujia are also growing steadily, pointing to their popularity as everyday snack options. Ketchup and Mixed Pickle have uneven trends with noticeable spikes. Meanwhile, Gooddot Proteiz and Nutrcharge items are becoming more popular, with occasional sharp rises especially in the case of Nutrcharge Woman. Overall, the SMA curves help clarify that even with some short-term fluctuations, most food products are becoming more popular over time, showing a shift in customer choices toward both convenience and healthier options.